

MASVINGO, Zimbabwe

WMO: 679750

Lat: 20.055S

Lon: 30.859E

Elev: 1096

StdP: 88.84

Time Zone: 2.00 (E02)

Period: 86-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.0	5.3	-4.3	3.0	20.3	-2.4	3.5	19.8	9.4	16.3	8.8	16.5	0.8	350	0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.8	33.7	17.6	32.7	17.6	31.5	17.4	21.5	26.7	21.1	26.1	20.7	25.5	4.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	17.1	22.6	19.9	16.7	22.3	19.5	16.3	22.1	68.4	26.7	66.7	26.4	65.1	25.5	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	8.1	7.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.6	22.9	22.4	21.7	19.4	16.6	14.4	14.1	16.8	19.9	22.0	22.9	22.5
	DBStd	4.05	1.97	1.84	2.14	2.34	2.32	2.45	2.11	2.89	2.94	3.29	2.75	2.34
	HDD10.0	1	0	0	0	0	0	0	1	0	0	0	0	0
	HDD18.3	430	1	0	2	14	65	120	131	66	17	9	3	2
	CDD10.0	3515	401	348	363	283	203	132	129	211	298	372	386	388
	CDD18.3	903	143	114	107	48	10	2	1	19	66	122	139	132
	CDH23.3	8623	1069	902	826	439	228	91	67	362	862	1378	1326	1074
	CDH26.7	2934	355	284	244	112	45	7	3	86	305	604	533	357
Wind	WSAvg	2.9	3.1	2.8	2.9	2.7	2.4	2.5	2.7	3.0	3.6	3.7	3.2	2.9
Precipitation	PrecAvg	604	126	105	62	25	11	6	3	4	10	27	78	146
	PrecMax	907	437	300	266	127	57	50	31	22	56	75	228	405
	PrecMin	224	2	4	1	0	0	0	0	0	0	0	7	23
	PrecStd	184	109	79	63	28	14	10	6	5	14	23	51	107
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.4	34.1	33.2	32.0	30.7	27.7	27.0	31.3	33.8	35.3	35.0	33.7
		MCWB	18.7	18.7	18.7	17.8	15.8	14.7	12.7	15.2	16.4	17.7	17.8	18.3
	2%	DB	31.7	32.0	31.1	29.7	28.1	25.8	25.3	29.4	31.9	33.4	33.2	32.0
		MCWB	18.8	18.9	18.6	17.9	15.1	13.7	12.6	14.7	15.5	16.8	17.5	18.8
	5%	DB	30.2	30.1	29.8	27.7	26.2	24.2	23.8	27.5	30.5	32.0	31.7	30.3
		MCWB	19.1	18.9	18.7	17.5	14.4	13.0	12.4	14.0	15.0	16.5	17.5	18.6
	10%	DB	28.7	28.7	28.2	25.9	24.3	22.4	22.1	25.6	28.9	30.2	29.8	28.8
		MCWB	19.4	19.1	18.9	17.2	14.1	12.5	12.1	13.4	14.7	16.4	17.7	18.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.2	21.9	21.9	20.5	17.3	16.1	15.0	17.2	18.8	20.1	21.2	21.7
		MCDB	27.3	27.0	27.2	26.4	24.7	24.0	21.5	26.9	26.5	27.8	28.4	27.0
	2%	WB	21.6	21.2	21.1	19.6	16.5	15.1	14.1	16.0	17.7	19.2	20.4	21.1
		MCDB	26.2	26.3	26.1	25.3	24.2	22.9	21.3	25.4	26.2	27.2	26.7	26.3
	5%	WB	21.1	20.7	20.5	19.0	15.7	14.2	13.4	15.1	16.8	18.6	19.7	20.6
		MCDB	25.5	25.7	25.2	24.1	23.3	21.6	20.7	24.4	25.7	26.7	25.5	25.7
	10%	WB	20.6	20.3	19.9	18.3	15.1	13.4	12.7	14.3	16.0	18.0	19.2	20.0
		MCDB	24.8	24.9	24.5	23.0	22.2	20.2	19.9	23.4	25.1	26.1	24.8	24.9
Mean Daily Temperature Range	5% DB	MDBR	9.8	10.4	10.6	11.8	14.6	15.1	14.6	15.3	14.8	13.2	11.4	10.6
		MCDBR	13.5	14.0	14.1	15.2	17.8	18.3	18.7	18.0	17.7	17.0	15.1	13.4
		MCWBR	3.9	4.1	4.5	5.2	6.9	7.7	8.1	6.4	5.9	5.1	4.3	3.7
	5% WB	MCDBR	8.8	9.2	10.0	11.3	14.2	15.1	14.3	15.2	13.8	12.9	9.8	10.1
		MCWBR	3.1	3.2	3.9	4.5	6.3	6.8	6.7	5.8	5.0	4.4	3.5	3.4
Clear-Sky Solar Irradiance	taub	0.353	0.349	0.338	0.310	0.285	0.281	0.289	0.326	0.400	0.400	0.379	0.366	
	taud	2.495	2.510	2.542	2.585	2.606	2.592	2.546	2.389	2.208	2.244	2.359	2.450	
	Ebn at Noon	990	983	970	960	950	935	936	928	894	922	958	978	
	Edh at Noon	116	113	105	93	84	82	88	112	144	146	132	122	
All-Sky Solar Radiation	RadAvg	6.42	6.31	5.65	5.18	4.81	4.23	4.31	5.33	5.99	6.43	6.25	6.12	
	RadStd	0.50	0.61	0.50	0.32	0.28	0.32	0.31	0.20	0.32	0.39	0.45	0.62	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page